

Solutions to Exercise Sheet 3: Groups and Lie Algebras

Exercise 1: Some properties of characters

Let G be a finite group, and let V and W be finite-dimensional representations of G over $\mathbb{C}$. Let

$$\rho_V : G \rightarrow GL(V), \quad \rho_W : G \rightarrow GL(W)$$

be the corresponding *representation* maps (group homomorphisms). Equivalently, a representation is an action of G on V such that for all $g, h \in G$ and $v \in V$,

$$g \cdot (h \cdot v) = (gh) \cdot v, \quad e \cdot v = v.$$

The character of V is the function

$$\chi_V : G \rightarrow \mathbb{C}, \quad \chi_V(g) = \text{tr}(\rho_V(g)).$$

Characters are class functions, meaning that they are constant on conjugacy classes.

We compute the characters of several naturally constructed representations.

1(a) Direct sum $V \oplus W$

The representation on $V \oplus W$ is given by

$$g \cdot (v, w) = (g \cdot v, g \cdot w).$$

In matrix form, after choosing bases of V and W , this action is block diagonal:

$$\rho_{V \oplus W}(g) = \begin{pmatrix} \rho_V(g) & 0 \\ 0 & \rho_W(g) \end{pmatrix}.$$

Taking traces gives

$$\chi_{V \oplus W}(g) = \text{tr}(\rho_V(g)) + \text{tr}(\rho_W(g)) = \chi_V(g) + \chi_W(g).$$

Thus

$$\boxed{\chi_{V \oplus W} = \chi_V + \chi_W.}$$

1(b) Tensor product $V \otimes W$

The tensor product representation is defined by

$$g \cdot (v \otimes w) = (g \cdot v) \otimes (g \cdot w).$$

In terms of matrices,

$$\rho_{V \otimes W}(g) = \rho_V(g) \otimes \rho_W(g).$$

We use the linear algebra identity

$$\mathrm{tr}(A \otimes B) = \mathrm{tr}(A) \mathrm{tr}(B).$$

Therefore

$$\chi_{V \otimes W}(g) = \mathrm{tr}(\rho_V(g) \otimes \rho_W(g)) = \mathrm{tr}(\rho_V(g)) \mathrm{tr}(\rho_W(g)).$$

Hence

$$\boxed{\chi_{V \otimes W}(g) = \chi_V(g) \chi_W(g).}$$

1(c) Dual representation V^*

The dual vector space is

$$V^* = \mathrm{Hom}_{\mathbb{C}}(V, \mathbb{C}).$$

The dual representation is defined by

$$(g \cdot \lambda)(v) = \lambda(g^{-1} \cdot v) \quad \text{for } \lambda \in V^*, v \in V.$$

Fix a basis of V , so that vectors are represented by column vectors and linear functionals by row vectors. Let $\rho_V(g) = A$. Then

$$g \cdot v = Av, \quad g^{-1} \cdot v = A^{-1}v.$$

By definition of the dual action,

$$(g \cdot \lambda)(v) = \lambda(g^{-1} \cdot v) = \lambda(A^{-1}v).$$

Since λ is a row vector, this equals

$$\lambda(A^{-1}v) = (\lambda A^{-1})v.$$

Thus

$$g \cdot \lambda = \lambda A^{-1}.$$

Passing to column vectors gives

$$\rho_{V^*}(g) = (A^{-1})^T = \rho_V(g^{-1})^T.$$

Taking traces gives

$$\chi_{V^*}(g) = \mathrm{tr}(\rho_V(g^{-1})^T) = \mathrm{tr}(\rho_V(g^{-1})) = \chi_V(g^{-1}).$$

For finite groups over $\mathbb{C}$, one may choose a G -invariant Hermitian inner product, making each $\rho_V(g)$ unitary. Hence

$$\chi_V(g^{-1}) = \overline{\chi_V(g)}.$$

Therefore

$$\boxed{\chi_{V^*}(g) = \chi_V(g^{-1}) = \overline{\chi_V(g)}.$$

1(d) Hom-space $\text{Hom}(V, W)$

There is a natural representation of G on $\text{Hom}(V, W)$ given by

$$(g \cdot T)(v) = g \cdot T(g^{-1} \cdot v).$$

Let

$$A = \rho_V(g), \quad B = \rho_W(g).$$

The action of g on $\text{Hom}(V, W)$ is therefore

$$T \mapsto BTA^{-1}.$$

As before, this means

$$\rho_{V \otimes W}(g) = (A^{-1})^T B = \rho_V(g^{-1})^T \rho_W(g),$$

and thus, over $\mathbb{C}$,

$$\boxed{\chi_{\text{Hom}(V, W)}(g) = \overline{\chi_V(g)} \chi_W(g).}$$

Similarly, there is a canonical vector-space isomorphism

$$\text{Hom}(V, W) \cong V^* \otimes W.$$

Therefore the character is

$$\chi_{\text{Hom}(V, W)}(g) = \chi_{V^*}(g) \chi_W(g).$$

1(e) Exterior square $\Lambda^2 V$

$$\Lambda^2 V = \text{span}\{v \wedge w : v, w \in V\}, \quad v \wedge w = v \otimes w - w \otimes v.$$

Let $A = \rho_V(g)$. Then

$$g \cdot (v \wedge w) = (Av) \wedge (Aw).$$

Choose a basis in which A is diagonal, with eigenvalues $\lambda_1, \dots, \lambda_n$. Then

$$g \cdot (e_i \wedge e_j) = \lambda_i \lambda_j (e_i \wedge e_j).$$

Thus

$$\chi_{\Lambda^2 V}(g) = \sum_{i < j} \lambda_i \lambda_j.$$

Using

$$\chi_V(g) = \sum_i \lambda_i, \quad \chi_V(g^2) = \sum_i \lambda_i^2,$$

we obtain

$$\chi_{\Lambda^2 V}(g) = \frac{1}{2} (\chi_V(g)^2 - \chi_V(g^2)).$$

1(f) Symmetric square $\text{Sym}^2 V$

Similarly, for $\text{Sym}^2 V$ which is the span of elements

$$v \odot w := v \otimes w + w \otimes v.$$

the eigenvalues of $\rho_{\text{Sym}^2 V}(g)$ are

$$\lambda_i \lambda_j \quad \text{for } 1 \leq i \leq j \leq n.$$

Hence

$$\chi_{\text{Sym}^2 V}(g) = \sum_{i \leq j} \lambda_i \lambda_j = \sum_i \lambda_i^2 + \sum_{i < j} \lambda_i \lambda_j.$$

Using the computation above,

$$\sum_{i \leq j} \lambda_i \lambda_j = \frac{1}{2} (\chi_V(g)^2 + \chi_V(g^2)).$$

Therefore

$$\chi_{\text{Sym}^2 V}(g) = \frac{1}{2} (\chi_V(g)^2 + \chi_V(g^2)).$$

Exercise 2: Quaternions

The quaternion group is

$$Q_8 = \{\pm 1, \pm i, \pm j, \pm k\},$$

with relations

$$\begin{aligned} i^2 &= j^2 = k^2 = -1, \\ ij &= k, \quad jk = i, \quad ki = j, \end{aligned}$$

and reversing the order changes the sign:

$$ji = -k, \quad kj = -i, \quad ik = -j.$$

2(a) The one-dimensional representations

A one-dimensional representation is a homomorphism

$$\rho : Q_8 \rightarrow \mathbb{C}^\times.$$

In this case, the character satisfies

$$\chi(g) = \rho(g),$$

so we may identify the representation with its character.

In any one-dimensional representation, i, j, k are sent to complex numbers, so they commute. Since

$$ij = k \quad \text{and} \quad ji = -k,$$

we must have

$$\chi(k) = \chi(-k).$$

Thus

$$\chi(-1) = 1.$$

The relations then imply

$$\chi(i)^2 = \chi(j)^2 = \chi(k)^2 = 1,$$

so $\chi(i), \chi(j), \chi(k) \in \{\pm 1\}$. Since $k = ij$, we have

$$\chi(k) = \chi(i)\chi(j).$$

Thus $\chi(i)$ and $\chi(j)$ may be chosen independently, and $\chi(k)$ is then determined.

The four one-dimensional characters are therefore:

	1	-1	$\pm i$	$\pm j$	$\pm k$
χ_1	1	1	1	1	1
χ_2	1	1	1	-1	-1
χ_3	1	1	-1	1	-1
χ_4	1	1	-1	-1	1

Here the notation $\pm i$ means that the character has the same value on i and $-i$, and similarly for $\pm j$ and $\pm k$.

2(b) The two-dimensional irreducible representation

The hint says to think of quantum mechanics. The relevant matrices are the Pauli matrices

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

They satisfy

$$\sigma_a^2 = I.$$

We define a representation $\rho : Q_8 \rightarrow GL_2(\mathbb{C})$ by

$$\rho(i) = i\sigma_1, \quad \rho(j) = i\sigma_2, \quad \rho(k) = -i\sigma_3.$$

Then

$$\rho(i)^2 = \rho(j)^2 = \rho(k)^2 = -I.$$

Moreover,

$$\rho(i)\rho(j) = (i\sigma_1)(i\sigma_2) = -\sigma_1\sigma_2 = -i\sigma_3 = \rho(k),$$

so the relation $ij = k$ is satisfied. Similarly one checks the other quaternion relations.

This gives a two-dimensional representation of Q_8 . Its character is

$$\chi_5(1) = 2, \quad \chi_5(-1) = -2,$$

and

$$\chi_5(\pm i) = \chi_5(\pm j) = \chi_5(\pm k) = 0,$$

since the Pauli matrices have trace zero.

This representation is irreducible. Indeed, if it had a nontrivial invariant subspace, then because it is two-dimensional, it would contain a common eigenline for all matrices $\rho(i), \rho(j), \rho(k)$. But $\rho(i)$

and $\rho(j)$ do not commute, and in fact there is no common eigenline preserved by both. Hence the representation is irreducible.

Suppose the representation were reducible. Since it is 2-dimensional, it would have a nonzero proper invariant subspace. Such a subspace must be 1-dimensional. Thus there would be a nonzero vector v such that

$$\rho(i)v = \alpha v, \quad \rho(j)v = \beta v$$

for some scalars $\alpha, \beta \in \mathbb{C}$.

Now use the quaternion relation

$$ij = -ji.$$

Applying both sides to v gives

$$\rho(i)\rho(j)v = -\rho(j)\rho(i)v.$$

Using the equations above, the left-hand side is

$$\rho(i)(\beta v) = \beta\rho(i)v = \beta\alpha v,$$

while the right-hand side is

$$-\rho(j)(\alpha v) = -\alpha\rho(j)v = -\alpha\beta v.$$

Hence

$$\alpha\beta v = -\alpha\beta v.$$

Since $v \neq 0$, this implies

$$2\alpha\beta = 0.$$

So

$$\alpha\beta = 0.$$

But $\rho(i)^2 = \rho(j)^2 = -I$, so neither $\rho(i)$ nor $\rho(j)$ can have eigenvalue 0. Therefore

$$\alpha \neq 0, \quad \beta \neq 0,$$

which contradicts $\alpha\beta = 0$.

Thus no nonzero proper invariant subspace exists, and the representation is irreducible.

2(c) Are these all irreducible representations?

For a finite group, the dimensions $d_1, \dots, d_r$ of the irreducible complex representations satisfy

$$\sum_{a=1}^r d_a^2 = |G|.$$

For Q_8 , we have already found four one-dimensional irreducible representations and one two-dimensional irreducible representation. Their squared dimensions sum to

$$1^2 + 1^2 + 1^2 + 1^2 + 2^2 = 4 + 4 = 8 = |Q_8|.$$

Therefore there is no room for any further irreducible representations. Hence these are all the irreducible complex representations of Q_8 .

2(d) Character table

The conjugacy classes of Q_8 are

$$\{1\}, \quad \{-1\}, \quad \{\pm i\}, \quad \{\pm j\}, \quad \{\pm k\}.$$

The character table is

	$\{1\}$	$\{-1\}$	$\{\pm i\}$	$\{\pm j\}$	$\{\pm k\}$
χ_1	1	1	1	1	1
χ_2	1	1	1	-1	-1
χ_3	1	1	-1	1	-1
χ_4	1	1	-1	-1	1
χ_5	2	-2	0	0	0

Exercise 3: The symmetric group S_4 and its representations

3(a) Conjugacy classes and Young diagrams

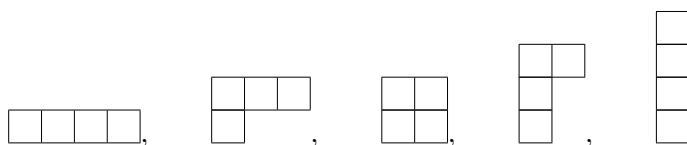
Conjugacy classes in S_n are determined by cycle type, equivalently by partitions of n . For S_4 , the partitions of 4 are

$$4, \quad 3 + 1, \quad 2 + 2, \quad 2 + 1 + 1, \quad 1 + 1 + 1 + 1.$$

These correspond to the following conjugacy classes:

partition	cycle type	example	class size
$1 + 1 + 1 + 1$	identity	$()$	1
$2 + 1 + 1$	transposition	(12)	6
$2 + 2$	double transposition	$(12)(34)$	3
$3 + 1$	3-cycle	(123)	8
4	4-cycle	(1234)	6

The Young diagrams corresponding to the partitions of 4 are



3(b) The two one-dimensional representations

A one-dimensional representation of S_4 is a homomorphism

$$S_4 \rightarrow \mathbb{C}^\times.$$

There are two such representations:

- the trivial representation, with character

$$\chi_{\text{triv}}(\sigma) = 1 \quad \text{for all } \sigma \in S_4;$$

- the sign representation, also called the alternating representation, with character

$$\chi_{\text{sgn}}(\sigma) = \text{sgn}(\sigma).$$

3(c) The standard representation

Let S_4 act on $\mathbb{R}^4$ by permuting coordinates:

$$\sigma \cdot (x_1, x_2, x_3, x_4) = (x_{\sigma^{-1}(1)}, x_{\sigma^{-1}(2)}, x_{\sigma^{-1}(3)}, x_{\sigma^{-1}(4)}).$$

This is the permutation representation.

The vector

$$(1, 1, 1, 1)$$

is fixed by all of S_4 , so it spans a one-dimensional trivial subrepresentation

$$\mathbb{R}(1, 1, 1, 1).$$

The complementary subspace

$$V = \{(x_1, x_2, x_3, x_4) \in \mathbb{R}^4 : x_1 + x_2 + x_3 + x_4 = 0\}$$

is invariant under coordinate permutations and has dimension 3.

Therefore

$$\mathbb{R}^4 \cong \mathbb{R}_{\text{triv}} \oplus V.$$

The representation V is the standard 3-dimensional irreducible representation of S_4 .

The character of the permutation representation is the number of fixed points of the permutation. Thus, subtracting the trivial character, the character of V is

$$\chi_V(\sigma) = \#\{\text{fixed points of } \sigma\} - 1.$$

On the conjugacy classes this gives

$$\chi_V = (3, 1, -1, 0, -1),$$

with the class order

$$(), \quad (12), \quad (12)(34), \quad (123), \quad (1234).$$

3(d) Remaining irreducible representations

Tensoring V with the sign representation gives another irreducible representation

$$V \otimes \text{sgn}.$$

Its character is

$$\chi_{V \otimes \text{sgn}}(\sigma) = \chi_V(\sigma) \text{sgn}(\sigma).$$

This gives

$$\chi_{V \otimes \text{sgn}} = (3, -1, -1, 0, 1).$$

The number of irreducible complex representations of a finite group equals the number of conjugacy classes. Since S_4 has 5 conjugacy classes, it has 5 irreducible representations.

We have already found:

$$\mathbf{1}, \quad \text{sgn}, \quad V, \quad V \otimes \text{sgn}.$$

Thus there is exactly one more irreducible representation.

Let its dimension be d . The sum of squares of dimensions equals $|S_4| = 24$:

$$1^2 + 1^2 + 3^2 + 3^2 + d^2 = 24.$$

Thus

$$20 + d^2 = 24, \quad d^2 = 4, \quad d = 2.$$

Therefore the remaining irreducible representation is 2-dimensional.

3(e) Character table of S_4

We use the class order

$$(), (12), (12)(34), (123), (1234).$$

The class sizes are

$$1, 6, 3, 8, 6.$$

The irreducible characters are:

	$()$	(12)	$(12)(34)$	(123)	(1234)
χ_{triv}	1	1	1	1	1
χ_{sgn}	1	-1	1	1	-1
χ_3	3	1	-1	0	-1
$\chi_{3'}$	3	-1	-1	0	1
χ_2	2	0	2	-1	0

Here χ_3 is the standard 3-dimensional representation, $\chi_{3'} = \chi_3 \otimes \chi_{\text{sgn}}$, and χ_2 is the remaining 2-dimensional irreducible representation.

To determine the remaining irreducible character χ_2 , write

$$\chi_2 = (2, a, b, c, d).$$

Using orthogonality with the known characters and the class sizes

$$1, 6, 3, 8, 6,$$

one obtains a system of linear equations. Solving it gives

$$a = 0, \quad b = 2, \quad c = -1, \quad d = 0.$$

Thus

$$\chi_2 = (2, 0, 2, -1, 0).$$

Exercise 4: Particle in a box

The Hilbert space is

$$\mathcal{H} = L^2([-l, l]^2),$$

with boundary condition

$$\psi|_{\partial[-l, l]^2} = 0.$$

The Hamiltonian is

$$H = -\frac{\hbar^2}{2m} (\partial_x^2 + \partial_y^2).$$

D4 symmetry

The square $[-l, l]^2$ is invariant under the dihedral group D_4 , the symmetry group of the square. This group contains rotations by multiples of $\pi/2$ and reflections across symmetry axes. If $g \in D_4$, define

$$(U_g \psi)(x, y) = \psi(g^{-1}(x, y)).$$

Because g preserves the square and its boundary, U_g preserves the boundary condition. Also, since g is an orthogonal transformation, the Laplacian is invariant:

$$\partial_x^2 + \partial_y^2$$

is unchanged under g . Hence

$$U_g H = H U_g.$$

Therefore the system has D_4 symmetry.

Solving the Schrödinger equation

We solve

$$-\frac{\hbar^2}{2m}(\partial_x^2 + \partial_y^2)\psi = E\psi$$

with Dirichlet boundary conditions

$$\psi(\pm l, y) = 0, \quad \psi(x, \pm l) = 0.$$

Use separation of variables:

$$\psi(x, y) = X(x)Y(y).$$

Then

$$-\frac{\hbar^2}{2m}(X''(x)Y(y) + X(x)Y''(y)) = EX(x)Y(y).$$

Dividing by $X(x)Y(y)$ gives

$$-\frac{\hbar^2}{2m}\left(\frac{X''(x)}{X(x)} + \frac{Y''(y)}{Y(y)}\right) = E.$$

Thus each variable satisfies a one-dimensional particle-in-a-box equation on $[-l, l]$ with zero boundary values.

The normalized one-dimensional eigenfunctions may be chosen as

$$\phi_n(t) = \frac{1}{\sqrt{l}} \sin\left(\frac{n\pi(t+l)}{2l}\right), \quad n = 1, 2, 3, \dots$$

They satisfy

$$\phi_n(-l) = \phi_n(l) = 0.$$

The corresponding one-dimensional energies are

$$\epsilon_n = \frac{\hbar^2}{2m} \left(\frac{n\pi}{2l}\right)^2.$$

Therefore the two-dimensional eigenfunctions are

$$\psi_{n_1, n_2}(x, y) = \phi_{n_1}(x)\phi_{n_2}(y), \quad n_1, n_2 \geq 1,$$

with energies

$$E_{n_1, n_2} = \frac{\hbar^2 \pi^2}{8ml^2} (n_1^2 + n_2^2).$$

A level transforming in the two-dimensional representation of D_4

If $n_1 \neq n_2$, then the two functions

$$\psi_{n_1, n_2}, \quad \psi_{n_2, n_1}$$

have the same energy. The D_4 action exchanges and changes signs among appropriate linear combinations of these functions. For example, take

$$(n_1, n_2) = (1, 2).$$

Then the eigenspace

$$E_{1,2} = E_{2,1}$$

is spanned by

$$\psi_{1,2}, \quad \psi_{2,1}.$$

We show that V is invariant under D_4 . For example, under a rotation exchanging x and y , one finds

$$(U_g \psi_{1,2})(x, y) = \psi_{1,2}(-y, x) = \pm \psi_{2,1}(x, y),$$

so the action maps V into itself. Similarly for all elements of D_4 . Thus D_4 acts on V , giving a representation

$$D_4 \rightarrow GL(V),$$

with $\dim V = 2$. This is the two-dimensional irreducible representation of D_4 . The energy is

$$E_{1,2} = \frac{\hbar^2 \pi^2}{8ml^2} (1^2 + 2^2) = \frac{5\hbar^2 \pi^2}{8ml^2}.$$

Lowest energy level that is not irreducible

Energy levels are determined by the integer

$$N = n_1^2 + n_2^2.$$

The first few values are:

(n_1, n_2)	$n_1^2 + n_2^2$	dimension of eigenspace
(1, 1)	2	1
(1, 2), (2, 1)	5	2
(2, 2)	8	1
(1, 3), (3, 1)	10	2
(2, 3), (3, 2)	13	2
(1, 4), (4, 1)	17	2
(3, 3)	18	1
(2, 4), (4, 2)	20	2
(3, 4), (4, 3)	25	2
(1, 5), (5, 1)	26	2
(1, 7), (7, 1), (5, 5)	50	3

The first time an energy eigenspace is not irreducible occurs when different unordered pairs give the same value of $n_1^2 + n_2^2$ or when a higher-dimensional eigenspace decomposes. A standard first example is

$$1^2 + 7^2 = 5^2 + 5^2 = 50.$$

Thus the energy level

$$E = \frac{\hbar^2 \pi^2}{8ml^2} \cdot 50 = \frac{25\hbar^2 \pi^2}{4ml^2}$$

has eigenspace spanned by

$$\psi_{1,7}, \quad \psi_{7,1}, \quad \psi_{5,5}.$$

The span of $\psi_{1,7}$ and $\psi_{7,1}$ gives a two-dimensional irreducible representation, while $\psi_{5,5}$ spans a one-dimensional representation. Hence this energy eigenspace is reducible and therefore does not transform irreducibly under D_4 .